24LC01B/02B

1K/2K 2.5V I²C™ Serial EEPROM

FEATURES

- Single supply with operation down to 2.5V
- Low power CMOS technology
  - 1 mA active current typical
  - 10 μA standby current typical at 5.5V
  - 5 μA standby current typical at 3.0V
- Organized as a single block of 128 bytes (128 x 8) -1K or 256 bytes (256 x 8) -2K
- 2-wire serial interface bus, I²C™ compatible
- Schmitt trigger inputs for noise suppression
- 100 kHz (E-temp.) and 400 kHz (C/I-temp.) compatibility
- Self-timed write cycle (including auto-erase)
- Page-write buffer for up to 8 bytes
- 2 ms typical write cycle time for page-write
- Hardware write protect for entire memory
- Can be operated as a serial ROM
- ESD protection > 3,000V
- 1,000,000 E/W cycles ensured
- Data retention > 200 years
- 8-pin DIP, SOIC, TSSOP or SOT-23* package
- Available for temperature ranges
  - Commercial (C): 0°C to +70°C
  - Industrial (I): -40°C to +85°C
  - Automotive (E): -40°C to +125°C

DESCRIPTION

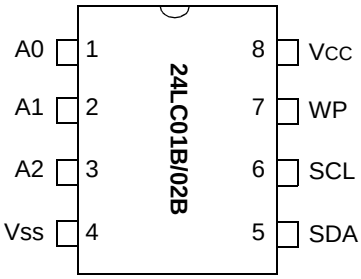
The Microchip Technology Inc. 24LC01B and 24LC02B are 1 Kbit and 2 Kbit Electrically Erasable PROMs (EEPROMs). The devices are organized as a single block of 128 x 8-bit or 256 x 8-bit memory with a 2-wire serial interface. Low voltage design permits operation down to 2.5 volts with a standby and active currents of only 5 μA and 1 mA respectively. The 24LC01B and 24LC02B also have page-write capability for up to 8 bytes of data. The 24LC01B and 24LC02B are available in the standard 8-pin DIP, 8-lead surface mount SOIC, MSOP and TSSOP packages. The SOT-23 package is available for the 24LC01B.

*Available for 24LC01B only

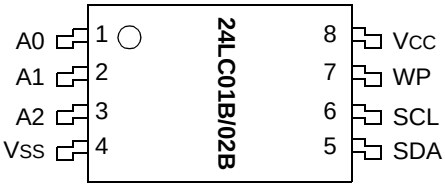
I²C is a trademark of Philips Corporation.

PACKAGE TYPES

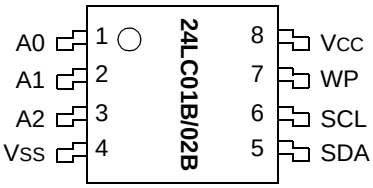
PDIP, SOIC



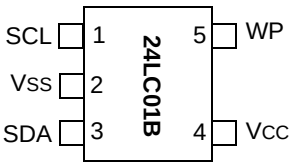
TSSOP



MSOP

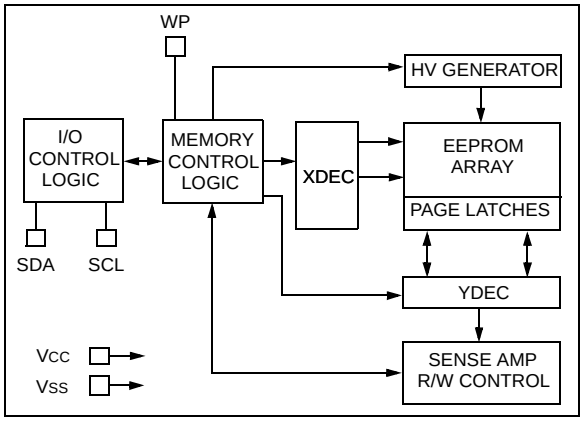


SOT-23*



Note: A0, A1 and A2 are not used
*Available for 24LC01B only

BLOCK DIAGRAM



1.0 ELECTRICAL CHARACTERISTICS

1.1 Maximum Ratings*

VCC.....7.0V
All inputs and outputs w.r.t. VSS .....-0.6V to VCC +1.0V
Storage temperature .....-65°C to +150°C
Ambient temp. with power applied .....-65°C to +125°C
Soldering temperature of leads (10 seconds) ..+300°C
ESD protection on all pins..... > 3 KV

***Notice:** Stresses above those listed under "Maximum ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

TABLE 1-1: PIN FUNCTION TABLE

Name	Function
VSS	Ground
SDA	Serial Address/Data I/O
SCL	Serial Clock
WP	Write Protect Input
VCC	+2.5V to 5.5V Power Supply
A0, A1, A2	**No Internal Connection**

TABLE 1-2: DC CHARACTERISTICS

VCC = +2.5V to +5.5V	**Commercial (C): TAMB = 0°C to +70°C Industrial (I): TAMB = -40°C to +85°C Automotive (E): TAMB = -40°C to +125°C**				
Parameter	**Symbol**	**Min.**	**Max.**	**Units**	**Conditions**
WP, SCL and SDA pins:					
High level input voltage	VIH	.7 VCC	—	V	—
Low level input voltage	VIL	—	.3 VCC	V	—
Hysteresis of Schmidt trigger inputs	VHYS	.05 VCC	—	V	**(Note)**
Low level output voltage	VOL	—	.40	V	IOL = 3.0 mA, VCC = 2.5V
Input leakage current	ILI	-10	10	µA	VIN = 0.1V to 5.5V
Output leakage current	ILO	-10	10	µA	VOUT = 0.1V to 5.5V
Pin capacitance (all inputs/outputs)	CIN, COUT	—	10	pF	VCC = 5.0V **(Note)** TAMB = 25°C, FCLK = 1 MHz
Operating current	ICC Write	—	3	mA	VCC = 5.5V, SCL = 400 kHz
	ICC Read	—	1	mA	—
Standby current	ICCS	—	30	µA	VCC = 3.0V, SDA = SCL = VCC
		—	100	µA	VCC = 5.5V, SDA = SCL = VCC WP = VSS

Note: This parameter is periodically sampled and not 100% tested.

FIGURE 1-1: BUS TIMING START/STOP

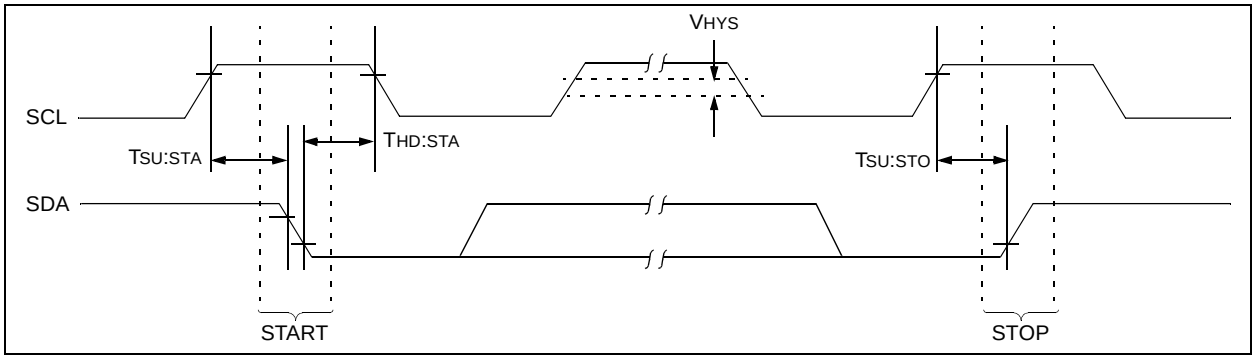


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TABLE 1-3: AC CHARACTERISTICS

VCC = +2.5V to 5.5V	Commercial (C): TAMB = 0°C to +70°C Industrial (I): TAMB = -40°C to +85°C Automotive (E): TAMB = -40°C to 125°C				
Parameter	**Symbol**	**Min**	**Max**	**Units**	**Conditions**
Clock frequency	FCLK	— —	400 100	kHz	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Clock high time	THIGH	600 4000	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Clock low time	TLOW	1300 4700	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
SDA and SCL rise time **(Note 1)**	TR	— —	300 1000	ns	4.5V ≤ VCC ≤ 5.5V **(Note 1)** 2.5V ≤ VCC ≤ 5.5V (E-temp. range) **(Note 1)**
SDA and SCL fall time	TF	—	300	ns	**(Note 1)**
START condition hold time	THD:STA	600 4000	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
START condition setup time	TSU:STA	600 4700	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Data input hold time	THD:DAT	0	—	ns	**(Note 2)**
Data input setup time	TSU:DAT	100 250	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp.range)
STOP condition setup time	TSU:STO	600 4000	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Output valid from clock **(Note 2)**	TAA	— —	900 3500	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Bus free time: Time the bus must be free before a new transmission can start	TBUF	1300 4700	— —	ns	4.5V ≤ VCC ≤ 5.5V 2.5V ≤ VCC ≤ 5.5V (E-temp. range)
Output fall time from VIH minimum to VIL maximum	TOF	20+0.1CB —	250 250	ns	4.5V ≤ VCC ≤ 5.5V **(Note 1)** 2.5V ≤ VCC ≤ 5.5V (E-temp. range) **(Note 1)**
Input filter spike suppression (SDA and SCL pins)	TSP	—	50	ns	**(Notes 1 and 3)**
Write cycle time (byte or page)	TWC	—	5	ms	—
Endurance		1M	—	cycles	25°C, VCC = 5.0V, Block Mode **(Note 4)**

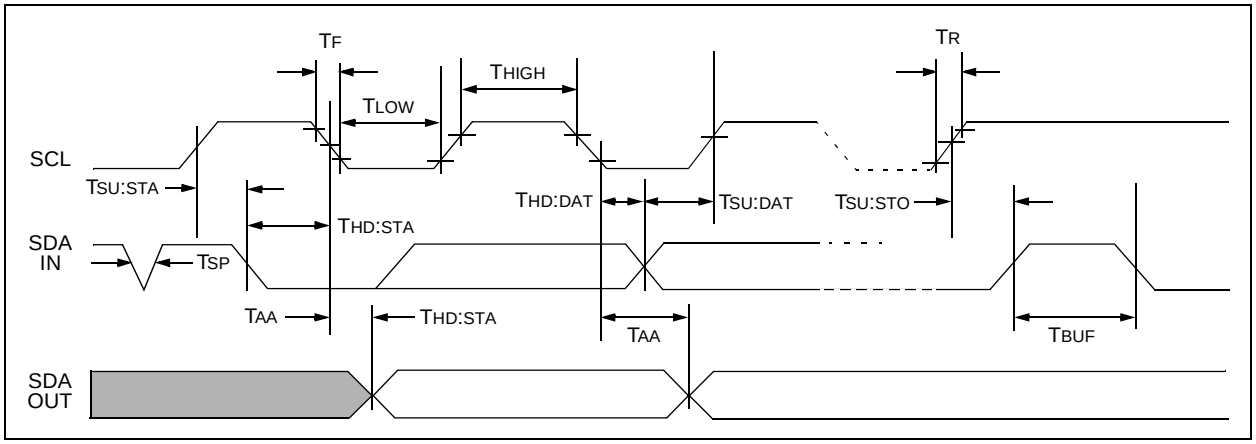
Note 1: Not 100% tested. CB = total capacitance of one bus line in pF.

2: As a transmitter, the device must provide an internal minimum delay time to bridge the undefined region (minimum 300 ns) of the falling edge of SCL to avoid unintended generation of START or STOP conditions.

3: The combined TSP and VHYS specifications are due to new Schmitt trigger inputs which provide improved noise spike suppression. This eliminates the need for a TI specification for standard operation.

4: This parameter is not tested but ensured by characterization. For endurance estimates in a specific application, please consult the Total Endurance Model which can be obtained on Microchip's website: www.microchip.com.

FIGURE 1-2: BUS TIMING DATA



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2.0 FUNCTIONAL DESCRIPTION

The 24LC01B/02B supports a bi-directional 2-wire bus and data transmission protocol. A device that sends data onto the bus is defined as transmitter and if receiving data, as receiver. The bus has to be controlled by a master device which generates the serial clock (SCL), controls the bus access and generates the START and STOP conditions, while the 24LC01B/02B works as slave. Both master and slave can operate as transmitter or receiver, but the master device determines which mode is activated.

3.0 BUS CHARACTERISTICS

The following **bus protocol** has been defined:

- Data transfer may be initiated only when the bus is not busy
- During data transfer, the data line must remain stable whenever the clock line is HIGH. Changes in the data line while the clock line is HIGH will be interpreted as a START or STOP condition

Accordingly, the following bus conditions have been defined (Figure 3-1).

3.1 Bus Not Busy (A)

Both data and clock lines remain HIGH.

3.2 Start Data Transfer (B)

A HIGH to LOW transition of the SDA line while the clock (SCL) is HIGH determines a START condition. All commands must be preceded by a START condition.

3.3 Stop Data Transfer (C)

A LOW to HIGH transition of the SDA line while the clock (SCL) is HIGH determines a STOP condition. All operations must be ended with a STOP condition.

3.4 Data Valid (D)

The state of the data line represents valid data when, after a START condition, the data line is stable for the duration of the HIGH period of the clock signal.

The data on the line must be changed during the LOW period of the clock signal. There is one clock pulse per bit of data.

Each data transfer is initiated with a START condition and terminated with a STOP condition. The number of the data bytes transferred between the START and STOP conditions is determined by the master device and is theoretically unlimited, although only the last sixteen will be stored when doing a write operation. When an overwrite does occur it will replace data in a first in first out fashion.

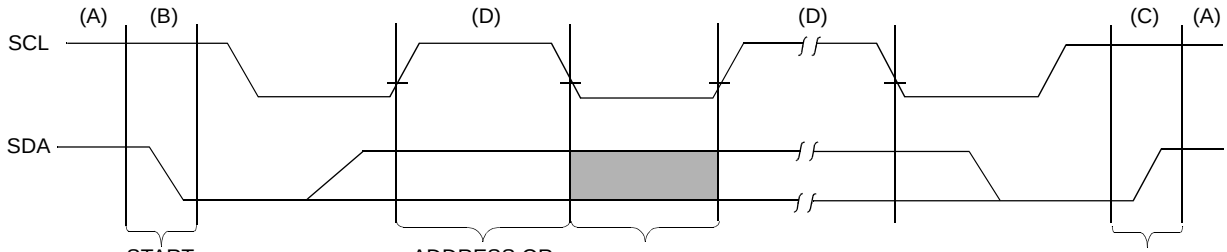
3.5 Acknowledge

Each receiving device, when addressed, is obliged to generate an acknowledge after the reception of each byte. The master device must generate an extra clock pulse which is associated with this acknowledge bit.

Note:	The 24LC01B/02B does not generate any acknowledge bits if an internal programming cycle is in progress.

The device that acknowledges has to pull down the SDA line during the acknowledge clock pulse in such a way that the SDA line is stable LOW during the HIGH period of the acknowledge related clock pulse. Of course, setup and hold times must be taken into account. A master must signal an end of data to the slave by not generating an acknowledge bit on the last byte that has been clocked out of the slave. In this case, the slave must leave the data line HIGH to enable the master to generate the STOP condition.

FIGURE 3-1: DATA TRANSFER SEQUENCE ON THE SERIAL BUS



17215.12 -96.544 T2 -060079 TcVA-6515(I)- 176.8(R)]T527.3

FIGURE 4-1: BYTE WRITE

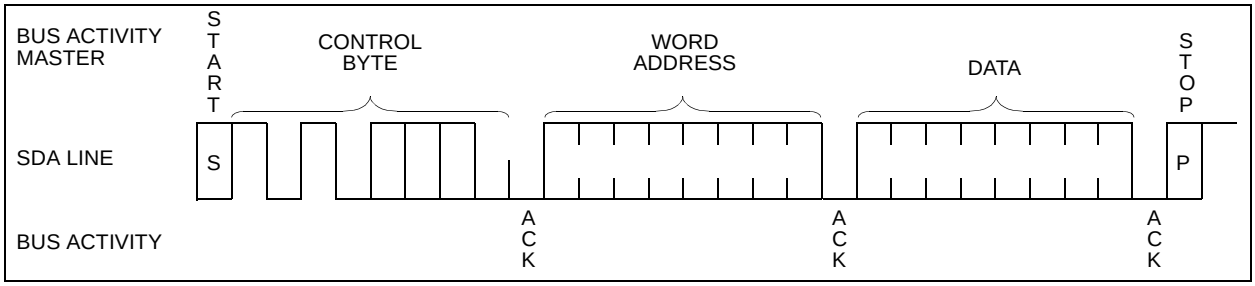
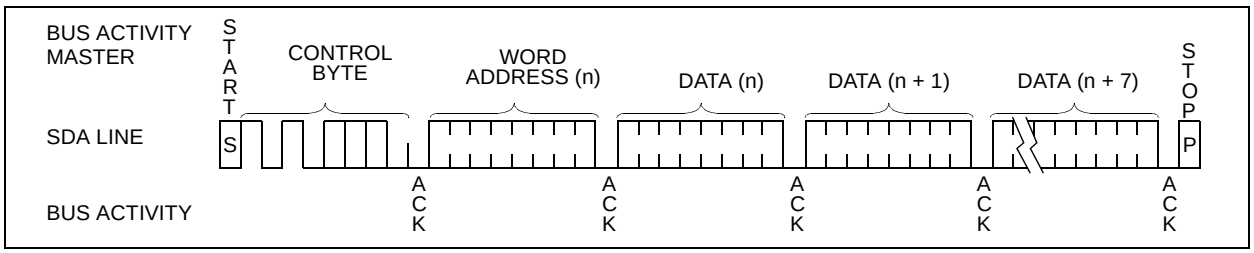


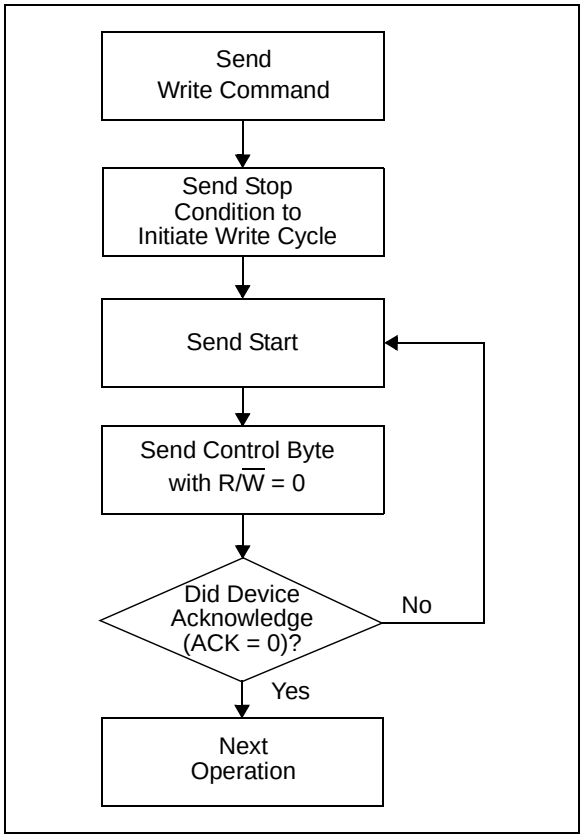
FIGURE 4-2: PAGE WRITE



5.0 ACKNOWLEDGE POLLING

Since the device will not acknowledge during a write cycle, this can be used to determine when the cycle is complete (this feature can be used to maximize bus throughput). Once the stop condition for a write command has been issued from the master, the device initiates the internally timed write cycle. ACK polling can be initiated immediately. This involves the master sending a start condition followed by the control byte for a write command (R/$\overline{W}$ = 0). If the device is still busy with the write cycle, then no ACK will be returned. If the cycle is complete, then the device will return the ACK and the master can then proceed with the next read or write command. See Figure 5-1 for flow diagram.

FIGURE 5-1: ACKNOWLEDGE POLLING FLOW



6.0 WRITE PROTECTION

The 24LC01B/02B can be used as a serial ROM when the WP pin is connected to VCC. Programming will be inhibited and the entire memory will be write-protected.

7.0 READ OPERATION

Read operations are initiated in the same way as write operations with the exception that the R/$\overline{W}$ bit of the slave address is set to '1'. There are three basic types of read operations: current address read, random read and sequential read.

7.1 Current Address Read

The 24LC01B/02B contains an address counter that maintains the address of the last word accessed, internally incremented by one. Therefore, if the previous access (either a read or write operation) was to address n, the next current address read operation would access data from address n + 1. Upon receipt of the slave address with R/$\overline{W}$ bit set to '1', the 24LC01B/02B issues an acknowledge and transmits the eight bit data word. The master will not acknowledge the transfer but does generate a stop condition and the 24LC01B/02B discontinues transmission (Figure 7-1).

7.2 Random Read

Random read operations allow the master to access any memory location in a random manner. To perform this type of read operation, first the word address must be set. This is done by sending the word address to the 24LC01B/02B as part of a write operation. After the word address is sent, the master generates a start condition following the acknowledge. This terminates the write operation, but not before the internal address pointer is set. Then the master issues the control byte again but with the R/$\overline{W}$ bit set to a '1'. The 24LC01B/02B will then issue an acknowledge and transmits the 8-bit data word. The master will not acknowledge the transfer but does generate a stop condition and the 24LC01B/02B discontinues transmission (Figure 7-2).

7.3 Sequential Read

Sequential reads are initiated in the same way as a random read except that after the 24LC01B/02B transmits the first data byte, the master issues an acknowledge as opposed to a stop condition in a random read. This directs the 24LC01B/02B to transmit the next sequentially addressed 8-bit word (Figure 7-3).

To provide sequential reads the 24LC01B/02B contains an internal address pointer which is incremented by one at the completion of each operation. This address pointer allows the entire memory contents to be serially read during one operation.

7.4 Noise Protection

The 24LC01B/02B employs a VCC threshold detector circuit which disables the internal erase/write logic if the VCC is below 1.5 volts at nominal conditions.

The SCL and SDA inputs have Schmitt trigger and filter circuits which suppress noise spikes to assure proper device operation even on a noisy bus.

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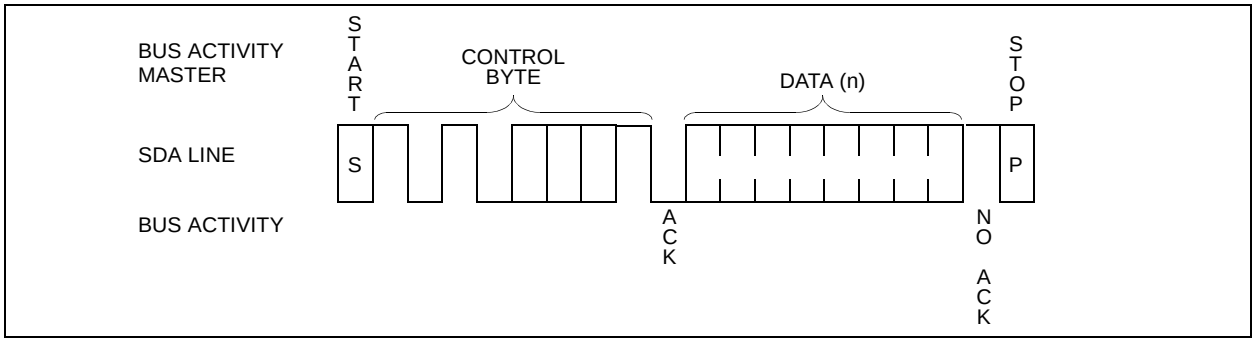
FIGURE 7-1: CURRENT ADDRESS READ



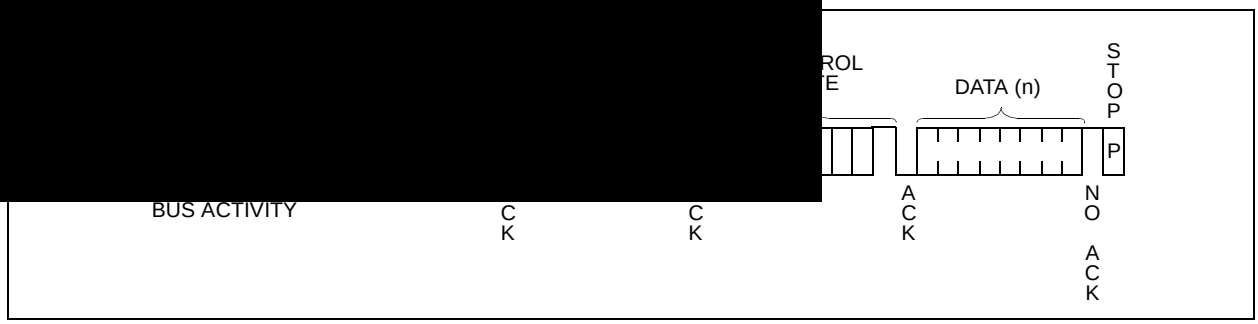


FIGURE 7-3: SEQUENTIAL READ

8.0 PIN DESCRIPTIONS

8.1 SDA Serial Address/Data Input/ Output

This is a bi-directional pin used to transfer addresses and data into and data out of the device. It is an open drain terminal. Therefore, the SDA bus requires a pull-up resistor to VCC (typical 10kΩ for 100 kHz, 2kΩ for 400 kHz).

For normal data transfer SDA is allowed to change only during SCL LOW. Changes during SCL HIGH are reserved for indicating the START and STOP conditions.

8.2 SCL Serial Clock

This input is used to synchronize the data transfer from and to the device.

8.3 WP

This pin must be connected to either VSS or VCC.

If tied to VSS, normal memory operation is enabled (read/write the entire memory).

If tied to VCC, WRITE operations are inhibited. The entire memory will be write-protected. Read operations are not affected.

This feature allows the user to use the 24LC01B/02B as a serial ROM when WP is enabled (tied to VCC).

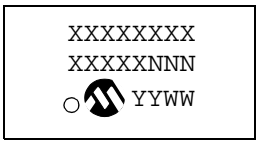
8.4 A0, A1, A2

These pins are not used by the 24LC01B/02B. They may be left floating or tied to either VSS or VCC.

9.0 PACKAGING INFORMATION

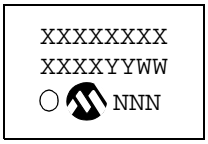
9.1 Package Marking Information

8-Lead PDIP (300 mil)



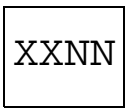
Example

8-Lead SOIC (150 mil)

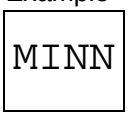


Example

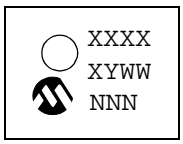
5-Lead SOT-23 (24LC01B only)



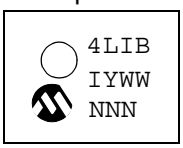
Example



8-Lead TSSOP



Example



8-Lead MSOP

Example

Legend:	XX...X	Customer specific information*
	Y	Year code (last digit of calendar year)
	YY	Year code (last 2 digits of calendar year)
	WW	Week code (week of January 1 is week ‘01’)
	NNN	Alphanumeric traceability code

Note: In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line thus limiting the number of available characters for customer specific information.

***** Standard PICmicro device marking consists of Microchip part number, year code, week code and traceability code. For PICmicro device marking beyond this, certain price adders apply. Please check with your Microchip Sales Office. For QTP devices, any special marking adders are included in QTP price.

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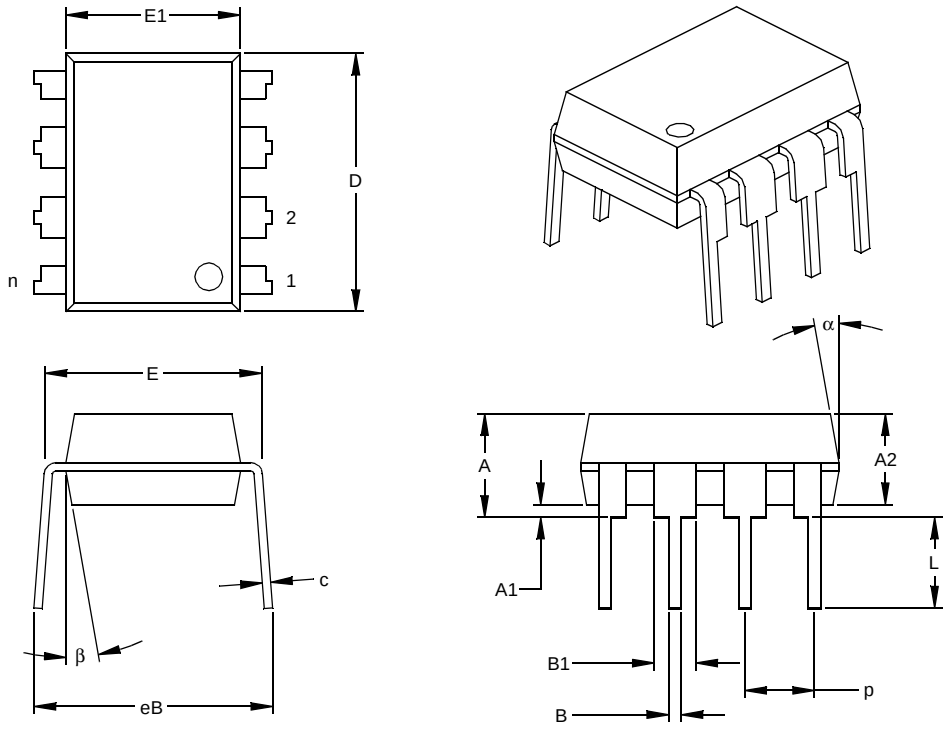
8-Lead Plastic Dual In-line (P) – 300 mil (PDIP)

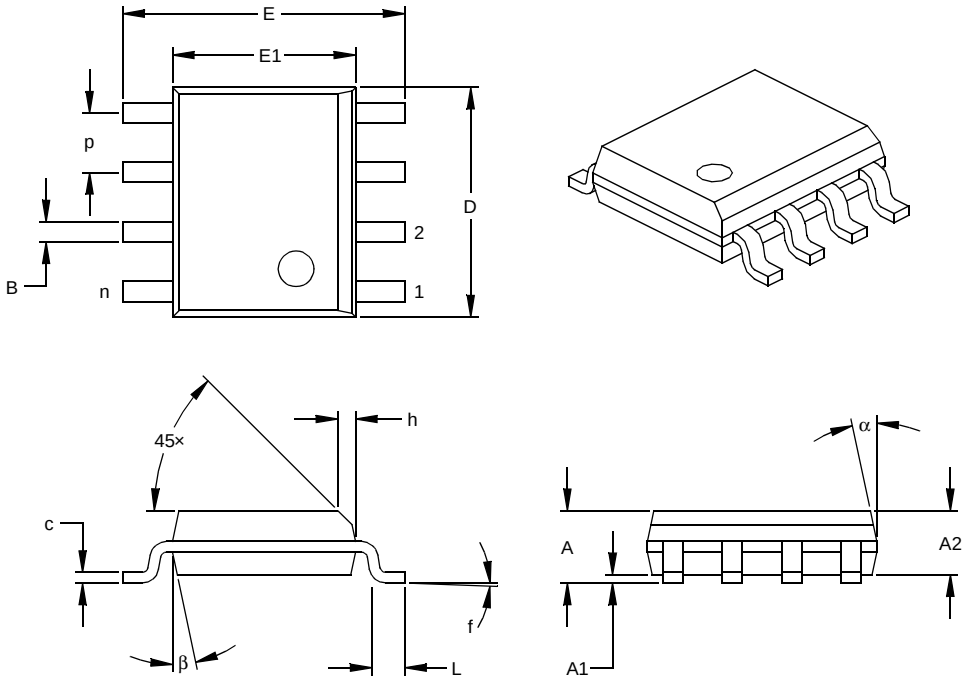


Units		INCHES*			MILLIMETERS		
Dimension Limits		MIN	NOM	MAX	MIN	NOM	MAX
Number of Pins	n		8			8	
Pitch	p		.100			2.54	
Top to Seating Plane	A	.140	.155	.170	3.56	3.94	4.32
Molded Package Thickness	A2	.115	.130	.145	2.92	3.30	3.68
Base to Seating Plane	A1	.015			0.38		
Shoulder to Shoulder Width	E	.300	.313	.325	7.62	7.94	8.26
Molded Package Width	E1	.240	.250	.260	6.10	6.35	6.60
Overall Length	D	.360	.373	.385	9.14	9.46	9.78
Tip to Seating Plane	L	.125	.130	.135	3.18	3.30	3.43
Lead Thickness	c	.008	.012	.015	0.20	0.29	0.38
Upper Lead Width	B1	.045	.058	.070	1.14	1.46	1.78
Lower Lead Width	B	.014	.018	.022	0.36	0.46	0.56
Overall Row Spacing §	eB	.310	.370	.430	7.87	9.40	10.92
Mold Draft Angle Top	α	5	10	15	5	10	15
Mold Draft Angle Bottom	β	5	10	15	5	10	15

* Controlling Parameter
§ Significant Characteristic

Notes:
Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .010” (0.254mm) per side.
JEDEC Equivalent: MS-001
Drawing No. C04-018

8-Lead Plastic Small Outline (SN) – Narrow, 150 mil (SOIC)



Units		INCHES*			MILLIMETERS		
Dimension Limits		MIN	NOM	MAX	MIN	NOM	MAX
Number of Pins	n		8			8	
Pitch	p		.050			1.27	
Overall Height	A	.053	.061	.069	1.35	1.55	1.75
Molded Package Thickness	A2	.052	.056	.061	1.32	1.42	1.55
Standoff §	A1	.004	.007	.010	0.10	0.18	0.25
Overall Width	E	.228	.237	.244	5.79	6.02	6.20
Molded Package Width	E1	.146	.154	.157	3.71	3.91	3.99
Overall Length	D	.189	.193	.197	4.80	4.90	5.00
Chamfer Distance	h	.010	.015	.020	0.25	0.38	0.51
Foot Length	L	.019	.025	.030	0.48	0.62	0.76
Foot Angle	f	0	4	8	0	4	8
Lead Thickness	c	.008	.009	.010	0.20	0.23	0.25
Lead Width	B	.013	.017	.020	0.33	0.42	0.51
Mold Draft Angle Top	α	0	12	15	0	12	15
Mold Draft Angle Bottom	β	0	12	15	0	12	15

* Controlling Parameter
§ Significant Characteristic

Notes:
Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .010” (0.254mm) per side.
JEDEC Equivalent: MS-012
Drawing No. C04-057

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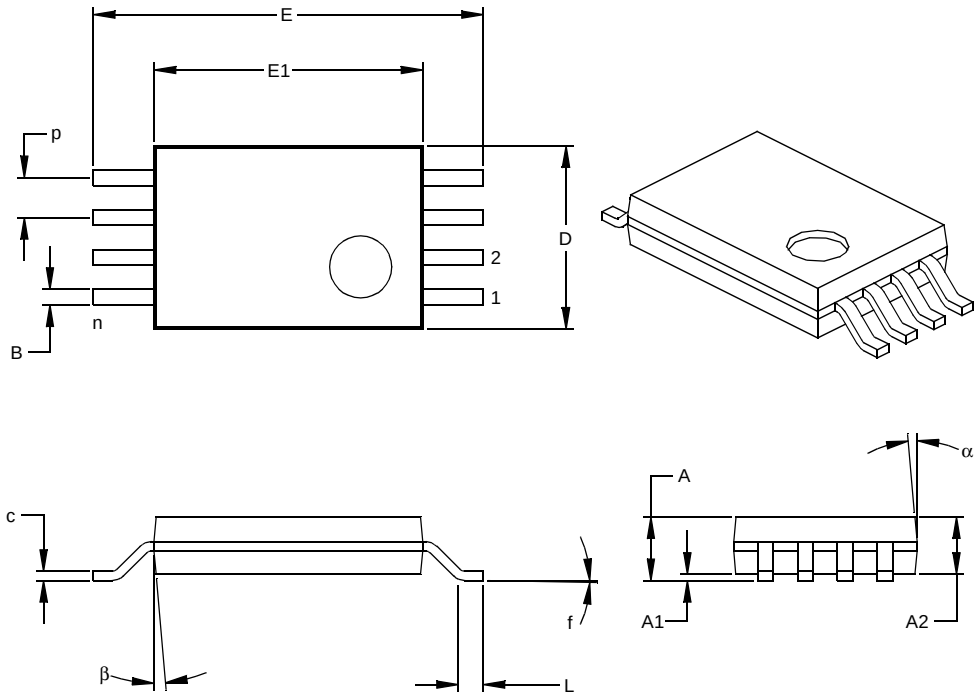
8-Lead Plastic Thin Shrink Small Outline (ST) – 4.4 mm (TSSOP)

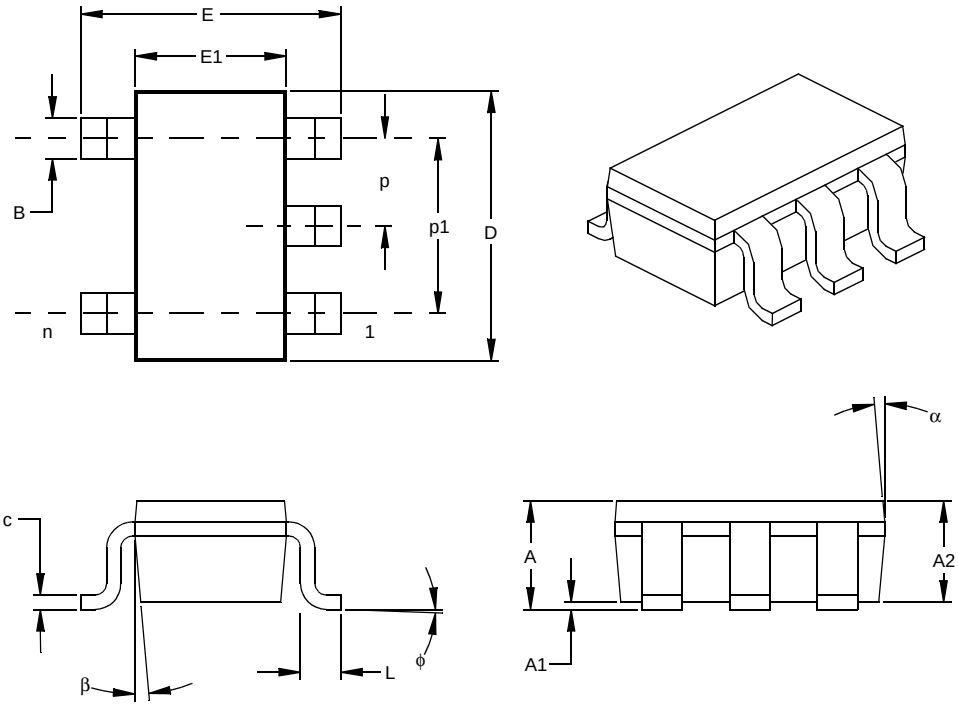


Units		INCHES			MILLIMETERS*		
Dimension Limits		MIN	NOM	MAX	MIN	NOM	MAX
Number of Pins	n		8			8	
Pitch	p		.026			0.65	
Overall Height	A			.043			1.10
Molded Package Thickness	A2	.033	.035	.037	0.85	0.90	0.95
Standoff §	A1	.002	.004	.006	0.05	0.10	0.15
Overall Width	E	.246	.251	.256	6.25	6.38	6.50
Molded Package Width	E1	.169	.173	.177	4.30	4.40	4.50
Molded Package Length	D	.114	.118	.122	2.90	3.00	3.10
Foot Length	L	.020	.024	.028	0.50	0.60	0.70
Foot Angle	f	0	4	8	0	4	8
Lead Thickness	c	.004	.006	.008	0.09	0.15	0.20
Lead Width	B	.007	.010	.012	0.19	0.25	0.30
Mold Draft Angle Top	α	0	5	10	0	5	10
Mold Draft Angle Bottom	β	0	5	10	0	5	10

* Controlling Parameter
§ Significant Characteristic

Notes:
Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .005” (0.127mm) per side.
JEDEC Equivalent: MO-153
Drawing No. C04-086

5-Lead Plastic Small Outline Transistor (OT) (SOT23)



Units		INCHES*			MILLIMETERS		
Dimension Limits		MIN	NOM	MAX	MIN	NOM	MAX
Number of Pins	n		5			5	
Pitch	p		.038			0.95	
Outside lead pitch (basic)	p1		.075			1.90	
Overall Height	A	.035	.046	.057	0.90	1.18	1.45
Molded Package Thickness	A2	.035	.043	.051	0.90	1.10	1.30
Standoff §	A1	.000	.003	.006	0.00	0.08	0.15
Overall Width	E	.102	.110	.118	2.60	2.80	3.00
Molded Package Width	E1	.059	.064	.069	1.50	1.63	1.75
Overall Length	D	.110	.116	.122	2.80	2.95	3.10
Foot Length	L	.014	.018	.022	0.35	0.45	0.55
Foot Angle	ϕ	0	5	10	0	5	10
Lead Thickness	c	.004	.006	.008	0.09	0.15	0.20
Lead Width	B	.014	.017	.020	0.35	0.43	0.50
Mold Draft Angle Top	α	0	5	10	0	5	10
Mold Draft Angle Bottom	β	0	5	10	0	5	10

* Controlling Parameter
§ Significant Characteristic

Notes:
Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .010” (0.254mm) per side.
JEDEC Equivalent: MO-178
Drawing No. C04-091

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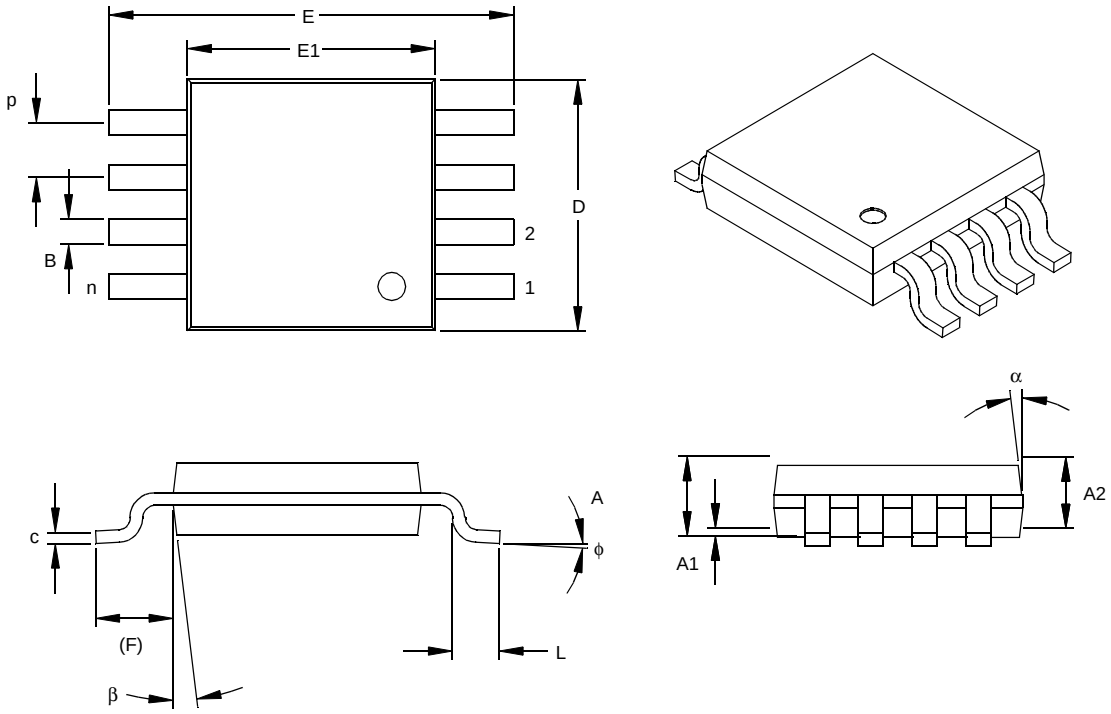
8-Lead Plastic Micro Small Outline Package (MS) (MSOP)



Units		INCHES			MILLIMETERS*		
Dimension Limits		MIN	NOM	MAX	MIN	NOM	MAX
Number of Pins	n		8				8
Pitch	p		.026			0.65	
Overall Height	A			.044			1.18
Molded Package Thickness	A2	.030	.034	.038	0.76	0.86	0.97
Standoff §	A1	.002		.006	0.05		0.15
Overall Width	E	.184	.193	.200	4.67	4.90	.5.08
Molded Package Width	E1	.114	.118	.122	2.90	3.00	3.10
Overall Length	D	.114	.118	.122	2.90	3.00	3.10
Foot Length	L	.016	.022	.028	0.40	0.55	0.70
Footprint (Reference)	F	.035	.037	.039	0.90	0.95	1.00
Foot Angle	φ	0		6	0		6
Lead Thickness	c	.004	.006	.008	0.10	0.15	0.20
Lead Width	B	.010	.012	.016	0.25	0.30	0.40
Mold Draft Angle Top	α		7			7	
Mold Draft Angle Bottom	β		7			7	

*Controlling Parameter
§ Significant Characteristic

Notes:

Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed .010" (0.254mm) per side.

Drawing No. C04-111

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	24LC01BT:	(Tape and Reel)	
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	24LC02BT:	(Tape and Reel)	
Temperature Range:	-	=	0°C to+70°C
	I	=	-40°C to+85°C
	E	=	-40°C to+125°C
Package:	P	=	Plastic DIP (300 mil body), 8-lead
	SN	=	Plastic SOIC (150 mil body), 8-lead
	OT	=	SOT-23, 5-lead (24LC01B only)
	ST	=	TSSOP, 8-lead
	MS	=	MSOP, 8-lead

Examples:

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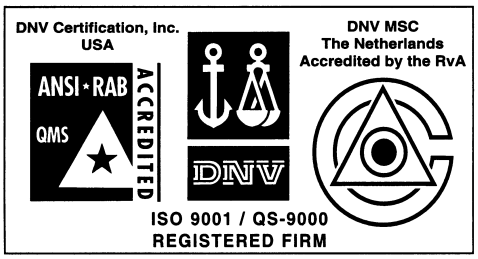

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New York
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Hauppauge, NY 11788
Tel: 631-273-5305 Fax: 631-273-5335

San Jose
Microchip Technology Inc.
2107 North First Street, Suite 590
San Jose, CA 95131
Tel: 408-436-7950 Fax: 408-436-7955

Toronto
6285 Northam Drive, Suite 108
Mississauga, Ontario L4V 1X5, Canada
Tel: 905-673-0699 Fax: 905-673-6509

ASIA/PACIFIC

Australia
Microchip Technology Australia Pty Ltd
Suite 22, 41 Rawson Street
Epping 2121, NSW
Australia
Tel: 61-2-9868-6733 Fax: 61-2-9868-6755

China - Beijing
Microchip Technology Beijing Office
Unit 915
New China Hong Kong Manhattan Bldg.
No. 6 Chaoyangmen Beidajie
Beijing, 100027, No. China
Tel: 86-10-85282100 Fax: 86-10-85282104

China - Shanghai
Microchip Technology Shanghai Office
Room 701, Bldg. B
Far East International Plaza
No. 317 Xian Xia Road
Shanghai, 200051
Tel: 86-21-6275-5700 Fax: 86-21-6275-5060

Hong Kong
Microchip Asia Pacific
RM 2101, Tower 2, Metroplaza
223 Hing Fong Road
Kwai Fong, N.T., Hong Kong
Tel: 852-2401-1200 Fax: 852-2401-3431

India
Microchip Technology Inc.
India Liaison Office
Divyasree Chambers
1 Floor, Wing A (A3/A4)
No. 11, O'Shaugnessey Road
Bangalore, 560 025, India
Tel: 91-80-2290061 Fax: 91-80-2290062

Japan
Microchip Technology Intl. Inc.
Benex S-1 6F
3-18-20, Shinyokohama
Kohoku-Ku, Yokohama-shi
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Tel: 81-45-471- 6166 Fax: 81-45-471-6122

ASIA/PACIFIC (continued)

Korea
Microchip Technology Korea
168-1, Youngbo Bldg. 3 Floor
Samsung-Dong, Kangnam-Ku
Seoul, Korea
Tel: 82-2-554-7200 Fax: 82-2-558-5934

Singapore
Microchip Technology Singapore Pte Ltd.
200 Middle Road
#07-02 Prime Centre
Singapore, 188980
Tel: 65-334-8870 Fax: 65-334-8850

Taiwan
Microchip Technology Taiwan
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EUROPE

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Germany
Arizona Microchip Technology GmbH
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Germany
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Arizona Microchip Technology SRL
Centro Direzionale Colleoni
Palazzo Taurus 1 V. Le Colleoni 1
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Milan, Italy
Tel: 39-039-65791-1 Fax: 39-039-6899883

United Kingdom
Arizona Microchip Technology Ltd.
505 Eskdale Road
Winnersh Triangle
Wokingham
Berkshire, England RG41 5TU
Tel: 44 118 921 5869 Fax: 44-118 921-5820

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